Energy Syst (2011) 2:313–324
DOI 10.1007/s12667-011-0038-5

ORIGINAL PAPER

Economic analysis of the $N - k$ power grid contingency selection and evaluation by graph algorithms and interdiction methods

Neng Fan · Hongsheng Xu · Feng Pan · Panos M. Pardalos

Received: 22 March 2011 / Accepted: 10 October 2011 / Published online: 27 October 2011
© Springer-Verlag 2011

Abstract Contingency analysis is important for providing information about the vulnerability of power grids. Many methods have been purposed to use topological structures of power grids for analyzing contingency states. Considering failures of buses and lines, we present and compare several graph methods for selecting contingencies in this paper. A new method, called critical node detection, is introduced for selecting contingencies consisting of failures on buses. Besides these methods, we include an interdiction model which provides the worst case contingency selection. Our measurement for contingency evaluation is to maximize the social benefit, or to minimize the generating and load shedding cost. Comparing with other measurements for contingency selection, our model is based on economic analysis and is reasonable for evaluating the selected contingency state. Additionally, a contingency consisting of both buses and lines is also studied.

Keywords Power grid · Contingency selection · Contingency analysis · Maximum-traffic · Betweenness · Critical node detection · Interdiction

Nomenclature

Indices and Index Sets

I: set of buses, including substations, generators and load consumers
Let $\mathbf{1}_i^g$ for $i \in I$ be the indicator such that $\mathbf{1}_i^g = 1$ if i is a generator, and $\mathbf{1}_i^g = 0$ otherwise

This work was supported in part by the Defense Threat Reduction Agency through the grant BRCALL08-A-2-0030.

N. Fan · F. Pan
Decision Applications Division, Los Alamos National Laboratory, Los Alamos, NM, USA

N. Fan (✉) · H. Xu · P.M. Pardalos
Center for Applied Optimization, Industrial and Systems Engineering, University of Florida, Gainesville, FL, USA
e-mail: andynfan@ufl.edu

$(i, j) \in L$: power lines, $i, j \in I$
$N = (I, L)$: an undirected network representing a power grid where I is the set of buses/nodes and L is the set of power lines/edges

Parameters
D_i: load at bus i
$\bar{F}_{ij}$: transmission capacity for line $(i, j) \in L$
$\bar{G}_i$: generation capacity at bus $i \in I$
b_{ij}: susceptance of line (i, j); b_{ij} is computed from the resistance and the reactance of line $(i, j) \in L$
h_i: generation cost at bus $i \in I$
r_i: penalty cost for load-shedding at $i \in I$
k: the number of components (buses or lines) to be failed

Decision Variables
g_i: power generation at node $i \in I$
f_{ij}: power flow on line $(i, j) \in L$
s_i: load shed at node $i \in I$
θ_i: phase angle at node $i \in I$
δ_i: $\delta_i = 1$ if the bus i is selected as one of k components for failures; $\delta_i = 0$ otherwise
σ_{ij}: $\sigma_{ij} = 1$ if the line (i, j) is selected as one of k components for failures; $\sigma_{ij} = 0$ otherwise

1 Introduction

Contingency analysis is a key function in the Energy Management System (EMS), which assesses the ability of power grids to sustain various combinations of power grid component failures based on state estimates. The contingency means a set of unexpected events happening within a short duration. The unexpected events can be failures of buses (generators, substations, etc) or transmission and distribution lines. In the past, the $N - 1$ contingency, or $N - 1$ reliability, analyzing the failure of one component (a bus or a line), has been an active research area [10, 15]. However, analysis of the $N - 1$ reliability is not sufficient for many real applications with multiple failures to discover the vulnerabilities of power grids.

The $N - k$ contingency analysis for failures of multiple components can reflect a larger variation of vulnerabilities of a power system. However, combinatorial number of contingency states imposes a substantial computational burden for analysis. Generally, contingency analysis consists of two steps: contingency selection and evaluation [14]. Contingency selection is to select a set of potential contingencies and to reduce the set of contingencies efficiently; contingency evaluation examines severe contingencies in details. Sometimes, vulnerability analysis can be used to find a set of small group of lines whose loss can cause a severe blackout.

Methods for contingency selection include automatic methods, online security analysis, ranking by approximate power flow solutions, concentric selection, sparse vector methods, genetic methods, hybrid methods and etc. These methods are briefly

reviewed in [7, 14]. Recently, probability analysis [6], power flow traffic [1, 19], limitations on generation and line capacity [5], and several other methods by studying the topology structures of power grids [11] are used for vulnerability analysis. In [11], Hines et al. reviewed and compared three vulnerability measures for contingency evaluation: characteristic path length, connectivity, blackout size. Additionally, they also compared the contingency selection methods, such as random failures, degree attack, maximum-traffic attack, minimum-traffic attack and betweenness attack, under three measures. The analysis considered the failures of buses only. In [11], they concluded that under the measure of characteristic path length, those buses selected by the node betweenness method are most crucial for the power grid; under the measure of connectivity loss, those buses selected by degree based method are most crucial; and under the measure of blackout size, those buses selected by maximum-traffic method are most crucial.

In this paper, besides methods compared in [11], we introduce the critical node detection method [2] to find buses who play important roles in a power grid. For failure on buses, we consider these graph algorithms: random, degree based, maximum-traffic, minimum-traffic, node betweenness, and critical node detection methods; For failures on lines, the graph algorithm considers the edge betweenness. After selecting contingencies by these graph algorithms, we evaluate the economic consequence of these contingency selection by using the direct current optimal power flow (OPF). The OPF minimizes the generating and load shedding cost or maximizes the social benefit in fact. The economic analysis for reliability or contingency analysis has been studied in some problems, for example, $N-1$ reliability in [15], and unit commitment [20]. In this paper, based on the OPF model whose objective is assumed to minimize the generating and load shedding cost, we use this objective as economic analysis to evaluate selected contingency states. By this analysis, we assume that under failures on some buses or lines, the power system is still operated in an optimized way. Especially, the load shedding cost can measure the unsatisfactory demand in case of failures in a system.

To compare the topology based methods, for the worst contingency in the OPF economic analysis model, we adapt the interdiction model introduced by [17, 18] to select k components for the worst failure and evaluate them. The interdiction model is to "identify critical sets of a power grid's components, e.g., generators, power lines, and transformers, by identifying maximally disruptive, coordinated attacks on a grid, which a terrorist group might undertake" [17]. By means of "maximally disruptive", the model is the worst case in economic analysis of the contingency.

The critical node detection method is formulated by a mixed integer linear program (MILP), and also the interdiction method by a mixed integer nonlinear program. Some heuristic method for critical node detection, node and edge betweenness are reviewed, while linearization techniques are proposed for nonlinear programs. All discussed graph algorithms and interdiction methods are compared on two IEEE test systems.

The rest of this paper is organized as follows: in Sect. 2 we present the linear programming OPF model for economic analysis; in Sect. 3, graph algorithms for contingency selection are reviewed and computational methods are also introduced; in Sect. 4, the interdiction method is introduced with formulations and methods; Sect. 5

presents the numerical experiments and comparisons in the objective of maximizing social benefit; and Sect. 6 concludes this paper.

2 OPF model for economic analysis

As discussed in Sect. 1, after contingency selection by graph algorithms, which decide the values of $\delta = (\delta_1, \ldots, \delta_{|I|}), \sigma = (\sigma_{ij} : i = 1, \ldots, |I|, j = i+1, \ldots, |I|)$, the OPF model is used for evaluating the selected contingencies. For a given power grid and given value of k (usually ranges from 2 to 10, [11]), each graph algorithm presents a contingency δ, σ. In this paper, we assume that on the tested system, each bus has no more than one generator, and a bus and its generator will always fail at the same time. In the following formulation (1)–(6), δ, σ are the known failed components in a power grid. To maximize the social benefit, we minimize the generating and load shedding cost $\sum_{i \in I}(h_i g_i + r_i s_i)$ in the OPF model in the power grids with failed buses or lines. The economic analysis model for power grids with failed components can be formulated as follows:

$$z(\delta, \sigma) = \min_{g,s,f,\theta} \sum_{i \in I}(h_i g_i + r_i s_i) \tag{1}$$

$$\text{s.t.} \quad f_{ij} = b_{ij}(\theta_i - \theta_j)(1-\delta_i)(1-\delta_j)(1-\sigma_{ij}), \quad \forall (i,j) \in L \tag{2}$$

$$-\bar{F}_{ij} \le f_{ij} \le \bar{F}_{ij}, \quad \forall (i,j) \in L \tag{3}$$

$$\sum_j f_{ji} + g_i = \sum_j f_{ij} + (D_i - s_i), \quad \forall i \in I \tag{4}$$

$$0 \le g_i \le \bar{G}_i(1-\delta_i), \quad \forall i \in I \tag{5}$$

$$0 \le s_i \le D_i, \quad \forall i \in I \tag{6}$$

For constraint (2) on line (i, j), it is an approximate active power flow on power line. If one ending bus of this line is failed (i.e., $\delta_i = 1$ or $\delta_j = 1$), or the line is failed (i.e., $\sigma_{ij} = 1$), the power flow on this line is 0. The constraint (3) ensures the maximum power flow on each line. The constraint (4) ensures the power balance at each bus, where the served demand at bus i is $(D_i - s_i)$. The constraint (5) limits the maximum generating output $\bar{G}_i(1-\delta_i)$. If the generator at bus i is failed (i.e., $\delta_i = 1$), no power is provided by this generator. The constraint (6) states the limitation of load shedding by the maximum load.

This formulation is a linear program with given values of δ, σ, and can be solved efficiently by simplex method in CPLEX.

3 Graph algorithms for contingency selection

In this section, graph algorithms are used for contingency selection of buses or lines, respectively.

3.1 Selection of k buses by critical node method and other methods

Besides using the contingency selection methods in [11], we add a new method, called critical node detection, for contingency selection of buses. For contingency selection on buses, other methods, such as power generation limits in [5] and ranking index [14], are also used for analysis.

In this section, we assume all failures, or attacks happen on buses. In the $N-k$ contingency analysis model, the number of failed buses is k, i.e., $\sum_{i\in I}\delta_i = k$.

The *critical node detection* problem (CNP), introduced in [2], is to detect a set of vertices in a graph whose deletion results in the graph having the minimum pairwise connectivity between the remaining vertices. We assume that k buses are failed with $\delta_i = 1$, and $\delta_i = 0$ for all other buses. In CNP, the set $\{i : \delta_i = 1\}$ of buses is to be selected. Let $x_{ij} = 1$ denote that buses i and j are in the same component after deletion of k buses with incident lines, and $x_{ij} = 0$ otherwise. The constraints of CNP include that the number of buses in the deleted subset is k, i.e., $\sum_{i:i\in I}\delta_i = k$, and all three pairs of three vertices have the relation that if two pairs are in the resulted graph, another pair is also in the resulted graph, i.e., $\max\{x_{ij} + x_{jt} - x_{it}, x_{ij} - x_{jt} + x_{it}, -x_{ij} + x_{jt} + x_{it}\} \le 1$. The relation between x_{ij} and δ_i can be expressed as $x_{ij} + \delta_i + \delta_j \ge 1$ under the objective of minimization. The objective of CNP is to minimize the pairwise connectivity between the remaining vertices. Therefore, the CNP for selecting k buses in a power grid $N = (I, L)$ can formulated as follows:

$$\min \sum_{i,j:v_i,v_j\in V} x_{ij} \tag{7}$$

$$\text{s.t.} \quad x_{ij} + \delta_i + \delta_j \ge 1, \quad \forall (i,j) \in L \tag{8}$$

$$x_{ij} + x_{jt} - x_{it} \le 1, \quad \forall i,j,t \in I \tag{9}$$

$$x_{ij} - x_{jt} + x_{it} \le 1, \quad \forall i,j,t \in I \tag{10}$$

$$-x_{ij} + x_{jt} + x_{it} \le 1, \quad \forall i,j,t \in I \tag{11}$$

$$\sum_{i:i\in I} \delta_i = k \tag{12}$$

$$\delta_i, x_{ij} \in \{0,1\}, \quad \forall i,j \in I \tag{13}$$

This problem is NP-hard [2] and the formulation (7)–(13) for CNP is a mixed integer linear program, which can be solved by CPLEX. In [2], heuristic methods based maximum independent set problem is proposed with computational complexity $O(|I|^2|L|)$. In this paper, for small IEEE test system, MILP by CPLEX is used and for large systems, heuristic methods are used. In (13), if i is required to satisfy $\delta_i = 0$ for $i \in \{i \in I : \mathbf{1}_i^g = 0\}$, all k failures are happening on generators. In this paper, we use heuristic methods proposed in [2] for numerical experiments.

The *random failure* [11], selects k buses for failure with equal probability. Thus, k buses from I are randomly selected and fixed $\delta_i = 1$. All other δ_is are set to be 0. The *degree* based method is to select k buses starting with the highest degree bus, till the kth highest degree, and set $\delta_i = 1$ for these corresponding k buses.

The *maximum-traffic* and *minimum-traffic* [1, 19] methods are based on the power flow of each bus:

$$T_i = |g_i - (D_i - s_i)| + \sum_{j:j\in I} |b_{ij}(\theta_i - \theta_j)|$$

where the term $|g_i - (D_i - s_i)|$ is the absolute value of net power injection into bus i by generators and load consumers, and the summation part is the sum of power flows in and out of bus i. The maximum-traffic method selects k buses with highest T_is, while the minimum-traffic method selects k buses with lowest T_is. In this paper, we use the OPF model (see Sect. 2) at normal state to find the traffic T_i on each bus.

The *node betweenness* method [1, 7] is to find k buses with highest node betweenness. The node betweenness [9] for bus i is defined as

$$C_B(i) = \sum_{s\neq i\neq t\in I} \frac{n_{st}(i)}{n_{st}}$$

where n_{st} is the number of shortest paths from s to t, and $n_{st}(i)$ is the number of shortest paths from s to t that pass through a bus i. To calculate the betweenness of all buses in a power grid, it needs calculating the shortest paths between all pairs of buses in a power grid. By modified Floyd-Warshall algorithm, and Johnson's algorithm for sparse graphs, the computational complexity can be $O(|I|^3)$ and $O(|I|^2 \log |I| + |I||L|)$, respectively. By using Brandes' algorithm [4], it can be more efficient with complexity $O(|I||L|)$.

3.2 Selection of k lines by edge betweenness

For failures on lines, we use edge betweenness [7] for contingency selection. The edge betweenness is adapted from node betweenness, and it can be expressed as

$$C_{(i,j)B} = \sum_{s,t\in I} \frac{n_{st}(i,j)}{n_{st}}$$

where n_{st} is the number of shortest paths from bus s to bus t, and $n_{st}(i,j)$ is the number of shortest paths from bus s to t that pass through line (i,j). The algorithms for finding the node betweenness can also be used to find the edge between for each line of a power grid. After obtaining k lines with highest edge betweenness, we set $\sigma_{ij} = 1$ for these lines and set $\sigma_{ij} = 0$ for all other lines.

Recently, studies of vulnerability for lines in power grid can be found in [3, 8, 16], where optimization methods are used to find small groups of lines whose failure can cause severe blackout. Additionally, contingency selection of lines is also studied by fitness function of line capacity [13], line capacity method [5] and etc.

4 The $N - k$ contingency selection by worst case interdiction analysis

In Sect. 3, by graph algorithms, we discussed the contingency selection for both buses and lines. However, except the maximum and minimum traffic methods for selecting

buses, all methods are from the prospective of graphs, or topology structures of power grids. In the following, we present an interdiction model for selecting the worst contingency case.

The interdiction model for power grid was proposed in [17, 18], where a contingency of maximal disruption to a power grid can be derived. The interdiction model has been used to model intelligent adversaries, e.g., terrorists. If $\{i : \delta_i = 1\}$ and $\{(i, j) : \sigma_{ij} = 1\}$ denote the sets of destroyed buses and lines, respectively, the constraint $\sum_{i\in I} \delta_i + \sum_{(i,j)\in L} \sigma_{ij} = k$ can be interpreted as a resource constraint on terrorists. The decision variables to maximally destroy a power grid include δ, σ. After the disruptive, the system operator solves the economic analysis model (1)–(6) by readjusting power generations, demands and power flows to minimize the consequence. The interdiction model is formulated as follows:

$$\max_{\delta,\sigma} \min_{g,s,f,\theta} \sum_{i\in I} (h_i g_i + r_i s_i) \tag{14}$$

$$\text{s.t.} \quad \text{Constraints (2)–(6)} \tag{15}$$

$$\sum_{i\in I} \delta_i + \sum_{(i,j)\in L} \sigma_{ij} = k \tag{16}$$

$$\delta_i, \sigma_{ij} \in \{0, 1\}, \quad \forall i \in I,\ (i, j) \in L \tag{17}$$

If the constraint $\sigma_{ij} = 0$ for all $(i, j) \in L$ is added to this model, all interdictions can only happen on buses. Similarly, if the constraint $\delta_i = 0$ is added, all interdictions can only happen on lines. Additionally, if we add $\sum_{i\in I} \delta_i = k_1$ and $\sum_{(i,j)\in L} \sigma_{ij} = k_2$, where $k_1 + k_2 = k$, the number of interdicted buses (or lines) are limited.

In this interdiction analysis model, for fixed δ, σ, the min part in (14) with constraints (15) construct the OPF model. Assume the associated dual variables for constraint (2) is α_{ij}, and β_{ij}, γ_{ij} for (3), η_i for (4), ζ_i for (5), μ_i for (6). Taking the duality of this part and combining unknown variables δ, σ, we can obtain the following equivalent formulation for the problem (14)–(17):

$$\max_{\delta,\sigma,\alpha,\beta,\gamma,\eta,\zeta,\mu} \sum_{(i,j)\in L} \bar{F}_{ij}(-\beta_{ij} + \gamma_{ij}) + \sum_{i\in I} D_i(\eta_i + \mu_i) + \sum_{i\in I} \bar{G}_i(1 - \delta_i)\zeta_i \tag{18}$$

$$\text{s.t.} \quad \eta_i + \zeta_i \le h_i \qquad \leftarrow g_i \tag{19}$$

$$\eta_i + \mu_i \le r_i \qquad \leftarrow s_i \tag{20}$$

$$\alpha_{ij} + \beta_{ij} + \gamma_{ij} - \eta_i + \eta_j = 0 \qquad \leftarrow f_{ij} \tag{21}$$

$$- \sum_{j:j>i} b_{ij}(1 - \delta_i)(1 - \delta_j)(1 - \sigma_{ij})\alpha_{ij}$$

$$+ \sum_{j:j<i} b_{ji}(1 - \delta_j)(1 - \delta_i)(1 - \sigma_{ji})\alpha_{ji} = 0 \qquad \leftarrow \theta_i \tag{22}$$

$$\beta_{ij} \ge 0,\ \gamma_{ij} \le 0,\ \zeta_i \le 0,\ \mu_i \le 0 \tag{23}$$

$$\text{Constraints (16), (17)} \tag{24}$$

where parameters on the right hand side denote corresponding variables of the original problem. This formulation is a mixed integer nonlinear program, and these terms, $\alpha_{ij}(1-\delta_i)(1-\delta_j)(1-\sigma_{ij})$, $(1-\delta_i)\zeta_i$ where $i = 1, \ldots, I$, $j = i+1, \ldots, I$, are nonlinear. To linearize them, we introduce some variables: $u_{ij} = (1-\delta_i)(1-\delta_j)(1-\sigma_{ij})$, $v_{ij} = \alpha_{ij}u_{ij}$ and $w_i = (1-\delta_i)\zeta_i$. The term $u_{ij} = (1-\delta_i)(1-\delta_j)(1-\sigma_{ij})$ can be linearized by

$$\begin{aligned} &u_{ij} \le 1-\delta_i, \qquad u_{ij} \le 1-\delta_j, \qquad u_{ij} \le 1-\sigma_{ij} \\ &u_{ij} \ge 1-\delta_i-\delta_j-\sigma_{ij}, \qquad u_{ij} \ge 0 \end{aligned} \tag{25}$$

the term $v_{ij} = \alpha_{ij}u_{ij}$ by

$$\begin{aligned} &v_{ij} \ge \alpha_{ij} - Ub\cdot(1-u_{ij}), \qquad v_{ij} \ge Lb\cdot u_{ij} \\ &v_{ij} \le \alpha_{ij} - Lb\cdot(1-u_{ij}), \qquad v_{ij} \le Ub\cdot u_{ij} \end{aligned} \tag{26}$$

and the term $w_i = (1-\delta_i)\zeta_i$ by

$$\begin{aligned} &w_i \ge \zeta_i - Ub\cdot\delta_i, \qquad w_i \ge Lb\cdot(1-\delta_i) \\ &w_i \le \zeta_i - Lb\cdot\delta_i, \qquad w_i \le Ub\cdot(1-\delta_i) \end{aligned} \tag{27}$$

where Lb, Ub are lower and upper bounds for α_{ij}, ζ_i and can be chosen as small enough or large enough constants, respectively.

By solving the problem (18)–(24) with linearizations (25), (26) and (27), which construct a mixed integer linear program, we can equivalently obtain solutions to the original problem (14)–(17). In this paper, we use CPLEX to solve the equivalent integer linear programming formulation.

5 Numerical experiments and comparisons

The linear program formulations for OPF and interdiction models are implemented in C++ and using CPLEX 12.1 via ILOG Concert Technology 2.9. The computational time for each experiment depends on the number of buses and lines of the tested system, and also the value of k. In this section, we test all proposed algorithms and methods for contingency selection and evaluation on IEEE-bus systems within a reasonable time (for most cases, less than 3000 seconds): IEEE-30-Bus system test case and RTS-96 test system [12].

In this section, we first consider $N-k$ contingency with k varying from 1 to 10. Bus and line failures are investigated separately for comparing costs. Finally, we consider failures of buses and lines in one contingency by interdiction model.

For experiments on IEEE-30-Bus System, the generating cost is not considered, i.e., setting $h_i = 0$ for $i \in I$. This system has 41 lines and 30 buses, 6 of which are generators and 21 of which are load buses. The relationship between load shedding cost and the number of failed buses under these graph algorithms and the interdiction

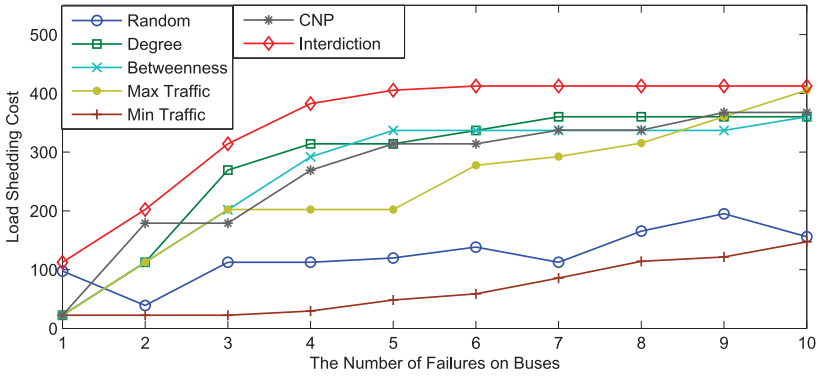


Fig. 1 Load shedding cost vs. failed buses (IEEE-30-Bus System)

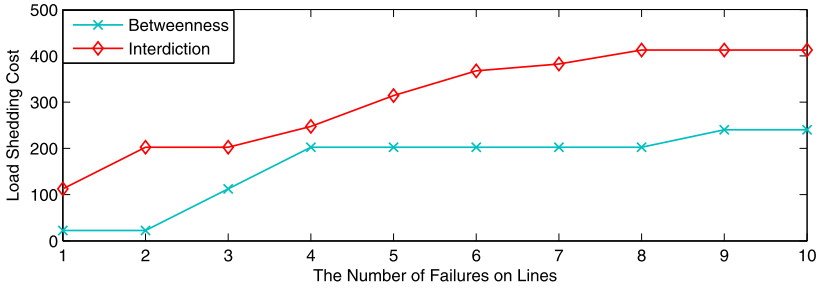


Fig. 2 Load shedding cost vs. failed lines (IEEE-30-Bus System)

method is shown in Fig. 1. From this figure, we can see, under the measure by minimizing load shedding, the interdiction method presents much worse power grid performance than other contingencies. The degree based, node betweenness, maximum-traffic and critical node detection (CNP) methods all select the crucial buses for this system, while the minimum-traffic cannot present crucial selection. Except the random failures, all costs proposed by other methods are monotonously increasing as well as the number k increases. Additionally, when $k \geq 6$ for the worst case interdiction method, load shedding costs for these cases are a constant. Checking these cases, we found that the whole power system is completely failed (no generation and no satisfied demand), and therefore, more failures on buses will not imply higher cost.

In Fig. 2 for IEEE-30-Bus System, we assume the contingency consisting only of lines, and assume load shedding is the only considered cost. Similarly, the interdiction method presents the worst case selection of lines as we expect. As the number k of failures is increasing, the load shedding cost increases as well.

As discussed in Sect. 4, the interdiction method can select the contingency consisting failures on both buses and lines. In Fig. 3, we show the cost of failures on both buses and lines. For given value of k, by adding $\sum_{i\in I}\delta_i = k_1$ and $\sum_{(i,j)\in L}\sigma_{ij} = k_2$ in the interdiction model, where $k_1 + k_2 = k$, we assume there are $k_1 = \lceil k/2 \rceil$ failed buses and $k_2 = \lfloor k/2 \rfloor$ failed lines. In this figure, we can find that the contingency con-

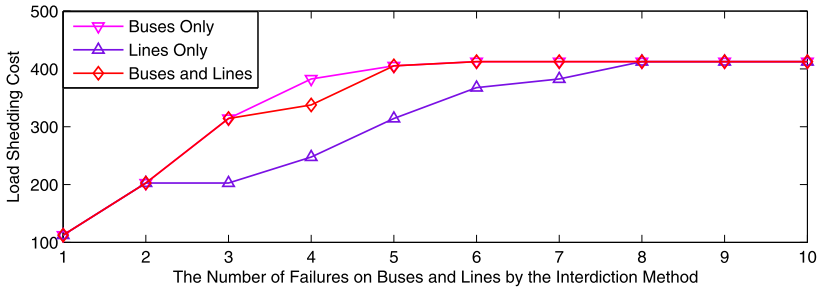


Fig. 3 Load shedding cost vs. failed buses/lines (IEEE-30-Bus System)

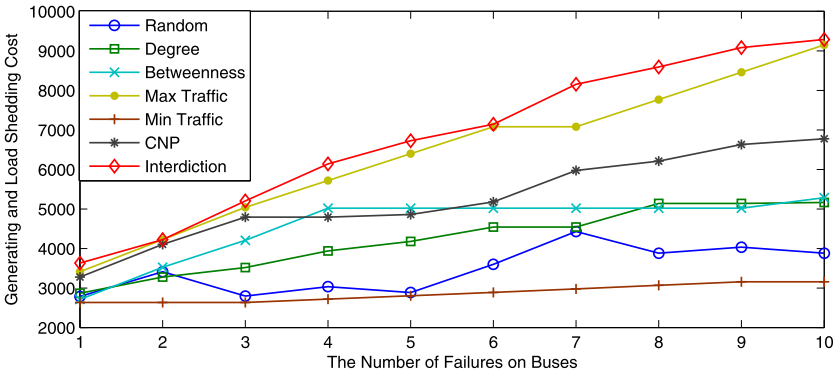


Fig. 4 Generating and load shedding cost vs. failed buses (RTS-96 System)

sisting of only buses has the most influence for the system. When k is large enough (e.g., $k \geq 6$ for cases of "buses only" and "buses and lines", $k \geq 8$ for the case of "lines only"), the whole system is completely failed. Additionally, if no assumptions made for k_1, k_2, according to our experiments, all failed components will be on buses. For the reason that buses are more important than nodes, one possible explanation is that our data sets assume that all lines have high capacity comparing those generations on buses.

To improve efficiency of the computation for MILP problems in large systems, for example, the interdiction method for RTS-96 System, the failed buses or lines with high appearances when k is small, can be fixed for decreasing the computational time when k is large. For experiments on RTS-96 System, both generating and load shedding costs are considered. This system has 73 buses and 108 combined lines. In Fig. 4, the relationship between generating and load shedding cost and the number of failed buses under graph algorithms and the interdiction method is presented. From this figure, we can see that, under the measure by minimizing generating and load shedding cost, the interdiction method presents the worst case selection of contingencies. The maximum-traffic method provides the second worst contingencies

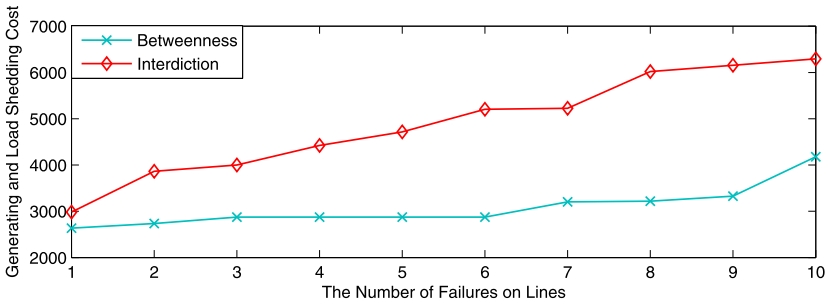


Fig. 5 Generating and load shedding cost vs. failed lines (RTS-96 System)

consistently, while the CNP method provides the third worst contingencies in most of values for k. Similarly, except the random failures, all costs proposed by other methods are monotonously increasing as well as the number k increases.

In Fig. 5 for RTS-96 System, assume the contingency consisting only of lines. Similarly, the interdiction method presents the worst case selection of lines as we expect. As the number k of failures is increasing, the generating and load shedding cost increases as well.

6 Conclusions

In this paper, we study the $N-k$ contingency selection and evaluation, by which the failures of multiple components can be selected and evaluated to reflect vulnerabilities of a power system. After reviewing some graph algorithms, we introduced the critical node detection method. These methods can find the contingencies consisting of failures only on buses, like generators, substations, etc. For contingency consisting of failures on power lines, the edge betweenness method is used. By maximizing the social benefit, or equivalently minimizing the generating and load shedding cost, we use economic analysis for evaluating selected contingencies. For the worst case contingency analysis, the interdiction method with limited k components' failures is used for both contingency selection and evaluation. Comparing with graph algorithms for contingency selection, the interdiction model can select contingencies consisting both buses and lines. By numerical experiments on two IEEE power grid test cases, we find that interdiction always selects the most crucial components, while the maximum-traffic method and critical node detection method select less crucial buses. For all cases, the minimum-traffic method finds the least crucial ones.

Future research directions include considering the security constraints in the OPF model, using alternating current power models, etc. In addition, using contingency analysis for finding vulnerabilities of power systems should provide suggestions for system planner and operator. To design a recovery plan for a power grid, or to design a warning system is also important.

References

1. Albert, R., Albert, I., Nakarado, G.L.: Structural vulnerability of the North American power grid. Phys. Rev. E **69**, 025103(R) (2004)
2. Arulselvan, A., Commander, C.W., Elefteriadou, L., Pardalos, P.M.: Detecting critical nodes in sparse graphs. Comput. Oper. Res. **36**(7), 2193–2200 (2009)
3. Bienstock, D., Verma, A.: The $N-k$ problem in power grids: new models, formulations and numerical experiments. SIAM J. Optim. **20**(5), 2352–2380 (2010)
4. Brande, U.: A faster algorithm for betweenness centrality. J. Math. Sociol. **25**, 163–177 (2001)
5. Carreras, B.A., Lynch, V.E., Dobso, I., Newman, D.E.: Critical points and transitions in an electric power transmission model for cascading failure blackouts. Chaos **12**(4), 985–994 (2002)
6. Chen, Q., McCalley, J.D.: Identifying high risk $N-k$ contingencies for online security assessment. IEEE Trans. Power Syst. **20**(2), 823–834 (2005)
7. Chen, Y., Jin, S., Chavarria-Miranda, D., Huang, Z.: Application of Cray XMT for power grid contingency selection. In: Proceedings of Cray User Group 2009, Atlanta, GA, May 2009
8. Donde, V., Lopez, V., Lesieutre, B., Pinar, A., Chao, Y., Meza, J.: Severe multiple contingency screening in electric power systems. IEEE Trans. Power Syst. **23**(2), 406–417 (2008)
9. Freeman, L.C.: A set of measures of centrality based on betweenness. Sociometry **40**, 35–41 (1977)
10. Hedman, K.W., Ferris, M.C., O'Neill, R.P., Fisher, E.B., Oren, S.S.: Co-optimization of generation unit commitment and transmission switching with $N-1$ reliability. IEEE Trans. Power Syst. **24**(2), 1052–1063 (2010)
11. Hines, P., Cotilla-Sanchez, E., Blumsack, S.: Do topological models provide good information about electricity infrastructure vulnerability? Chaos **20**, 033122 (2010)
12. IEEE reliability test data [online]. Available at http://www.ee.washington.edu/research/pstca/
13. Li, F., Chegu, A.: High order contingency selection using particle swarm optimization and Tabu search. In: Proceedings of 15th International Conference on Intelligent System Applications to Power Systems, 2009, ISAP '09 (2009)
14. Mori, H., Goto, Y.: A tabu search based approach to $(N-k)$ static contingency selection in power systems. In: Proceedings of 2001 IEEE International Conference on Systems, Man, and Cybernetics, vol. 3, pp. 1954–1959 (2001)
15. O'Neill, R.P., Hedman, K.W., Krall, E.A., Papavasiliou, A., Oren, S.S.: Economic analysis of the ISOs multi-period $N-1$ reliable unit commitment and transmission switching problem using duality concepts. Energy Syst. **1**(2), 165–195 (2010)
16. Pinar, A., Meza, J., Donde, V., Lesieutre, B.: Optimization strategies for the vulnerability analysis of the electric power grid. SIAM J. Optim. **20**(4), 1786–1810 (2010)
17. Salmerón, J., Wood, K., Baldick, R.: Analysis of electric grid security under terrorist threat. IEEE Trans. Power Syst. **19**(2), 905–912 (2004)
18. Salmerón, J., Wood, K., Baldick, R.: Worst-case interdiction analysis of large-scale electric power grids. IEEE Trans. Power Syst. **24**(1), 96–104 (2009)
19. Wang, J.-W., Rong, L.-L.: Cascade-based attack vulnerability on the US power grid. Saf. Sci. **47**, 1332–1336 (2009)
20. Wu, L., Shahidehpour, M., Li, T.: Cost of reliability analysis based on stochastic unit commitment. IEEE Trans. Power Syst. **23**(3), 1364–1374 (2008)